

Hough Circle Transform

Formula for radius:

$$r = \sqrt{(x_p - x_c)^2 + (y_p - y_c)^2}$$

For $c_1 = (2, 4)$:

• $P_1 = (5, 4)$

$$r = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3$$

• $P_2 = (7, 6)$

$$r = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4} = \sqrt{29} \approx 5.39$$

For $c_2 = (4, 3)$:

• $P_1 = (5, 4)$

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2} \approx 1.41$$

• $P_2 = (7, 6)$

$$r = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.24$$

Result:

<u>Center</u>	<u>P_1 Radius</u>	<u>P_2 radius</u>	<u>Matching radius</u>
$c_1(2, 4)$	3	5.39	0
$c_2(4, 3)$	1.41	4.24	0

② Given:

$$P = (8, 6)$$

$$C = (4, 3)$$

$$\text{Radius} = \sqrt{(x_P - x_C)^2 + (y_P - y_C)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{16+9}$$

$$= \sqrt{25}$$

$$= 5$$

$$\boxed{r = 5}$$

③ Given:

$$C = (5, 4)$$

$$P = (8, 8)$$

$$\text{Radius: } \sqrt{(x_P - x_C)^2 + (y_P - y_C)^2}$$

$$= \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5$$

$$\therefore \boxed{r = 5}$$